COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Tutorial Sheet: Unit 2 – Sessions 1, 2 & 3 (Sections C, D & E)
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Section C – Diagram-Based Questions

• Basic Functional Units Diagram (described): Input Unit and Output Unit attached to Memory Unit; Memory Unit connected to CPU containing ALU + Control Unit; Control Unit sends control signals to all units; arrows show data flow Input → Memory → ALU → Memory → Output.

• Single Bus Structure Diagram (described): CPU, Memory and all I/O devices each interfaced to one common bus; the bus is subdivided into Data Bus, Address Bus and Control Bus lines.

Section D – Conceptual/Application Questions

• Data and control flow during instruction execution: PC supplies the instruction address; the instruction is fetched from memory into IR; the Control Unit decodes it and issues control signals; operands are fetched using the Address Bus; the ALU performs the operation; the result is stored back in memory or a register; PC is updated for the next instruction.

• Single Bus System Instruction Flow: PC → MAR → memory fetch; MDR → IR; Control Unit executes; any operand is fetched over the same shared bus; only one operation at a time is possible because the bus is shared.

• Add R1, R2, R3 execution: fetch instruction; decode (identify ADD); fetch contents of R2 and R3; ALU computes R2 + R3; store the result in R1.

Section E – Challenge Task / Group Activity

• Discussion point: We prefer a multiple bus structure for high-performance systems because it allows parallel data transfer, reduces bottlenecks and improves CPU efficiency. However, it is costlier and more complex to manage, requiring hardware arbiters and more interconnections.